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RBP-J regulates homeostasis and function of circulating Ly6C^{lo} monocytes

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Abstract

Summary

Notch-RBP-J signaling plays an essential role in maintenance of myeloid homeostasis. However, its role in monocyte cell fate decisions is not fully understood. Here we showed that conditional deletion of transcription factor RBP-J in myeloid cells resulted in marked accumulation of blood Ly6C^{lo} monocytes that highly expressed chemokine receptor CCR2. Bone marrow transplantation and parabiosis experiments revealed a cell intrinsic requirement of RBP-J for controlling blood Ly6C^{lo}CCR2^{hi} monocytes. RBP-J-deficient Ly6C^{lo} monocytes exhibited enhanced capacity competing with wildtype counterparts in blood circulation. In accordance with alterations of circulating monocytes, RBP-J deficiency led to markedly increased population of lung tissues with Ly6C^{lo} monocytes and CD16.2^{+} interstitial macrophages. Furthermore, RBP-J deficiency-associated phenotypes could be genetically corrected by further deleting *Ccr2* in myeloid cells. These results demonstrate that RBP-J functions as a crucial regulator of blood Ly6C^{lo} monocytes and thus derived lung-resident myeloid populations, at least in part through regulation of CCR2.

eLife assessment

This study presents a **valuable** examination into the role Notch-RBP-J signaling in regulating monocyte subset homeostasis. The data were collected and analyzed using **solid** and validated methodology and can be used as a starting point for exploring the mechanisms involved in RBP-J signaling in non-classical monocytes.

Introduction

Monocytes are integral components of the mononuclear phagocyte system (MPS) that develop from monocyte precursors in the bone marrow[1]. Several functionally and phenotypically distinct subsets of blood monocytes have been defined on the basis of expression of surface markers[(2)-(5)]. In mice, Ly6C^{hi} monocytes (also called inflammatory or classical monocytes) characterized by Ly6C^{hi}CCR2^{hi}CX3CR1^{lo} exhibit a short half-life, and are recruited to tissue and differentiate into macrophages and dendritic cells[(6)]. Ly6C^{hi} monocytes are analogous to CD14^{+} human monocytes based on gene expression profiling[(7)]. A second subtype of monocytes is called Ly6C^{lo} monocytes (also termed patrolling monocytes or non-classical monocytes), which express high level of CX3CR1 and low level of CCR2, equivalent to CD14^{lo}CD16^{+} human monocytes. Ly6C^{lo} monocytes are regarded as blood-resident macrophages that patrol blood vessels and scavenge microparticles attached to the endothelium under physiological conditions, and exhibit a long half-life in the steady-state[(8)-(10)]. Ly6C^{lo} monocytes may exert a protective effect by suppressing atherosclerosis in mice, while high levels of nonclassical monocytes have been associated with more advanced vascular dysfunction and oxidative stress in patients with coronary artery disease[(11)-(13)]. In the inflammatory settings, Ly6C^{lo} monocytes often show anti-inflammatory properties, yet also exhibit a proinflammatory role under certain circumstances[(5),(9),(14)-(18)]. In addition, Ly6C^{lo} monocytes have demonstrated protective functions in tumorigenesis, such as engulfing tumor materials to prevent cancer metastasis, activating and recruiting NK cells to the lungs[(19),(20)]. While the functions of Ly6C^{lo} monocytes are complex and sometimes contradictory depending on the animal models and experimental approaches, it is clear that they play a crucial role in both health and disease.

The generation and maintenance of Ly6C^{lo} monocytes are regulated by several transcription factors, including Nr4a1 and CEBPβ[(21),(22)]. The colony-stimulating factor 1 receptor (CSF1R) signaling pathway is important for the generation and survival of Ly6C^{lo} monocytes, and neutralizing antibodies against CSF1R can reduce their numbers[(23)]. Interestingly, recent studies have suggested that different subsets of monocytes may be supported by distinct cellular sources of CSF-1 within bone marrow niches, in which targeted deletion of *Csf1* from sinusoidal endothelial cells selectively reduces Ly6C^{lo} monocytes but not Ly6C^{hi} monocytes[(24)]. LAIR1 has been shown to be activated by stromal protein Colec12, and LAIR1 deficiency leads to aberrant proliferation and apoptosis in bone marrow non-classical monocytes[(25)]. These findings further underscore the complexity of the mechanisms that regulate Ly6C^{lo} monocyte, and suggest that multiple factors are involved in this process.

Recombination signal-binding protein for immunoglobulin kappa J region (RBP-J; also named CSL in humans) is commonly known as the master nuclear mediator of canonical Notch signaling[(26)]. In the absence of Notch intracellular domain, RBP-J associates with co-repressor proteins to repress transcription of downstream target genes. In the immune system, the best studied functions of Notch signaling are its roles in regulating the development of lymphocytes, such as T cells and marginal zone B cells[(26)]. Notch signaling also has been reported to regulate the differentiation and function of myeloid cells including granulocyte/monocyte progenitors, osteoclasts and dendritic cells [(27)-(30)]. In osteoclasts, RBP-J represses osteoclastogenesis, particularly under inflammatory conditions[(30)]. In dendritic cells, Notch-RBP-J signaling controls the maintenance of CD8^{-} DC[(28)]. In addition, Notch2-RBP-J pathway is essential for development of CD8^{-}ESAM^{+} DC in the spleen and CD103^{+}CD11b^{+} DC in the lamina propria of the intestine[(29)]. However, the role of Notch-RBP-J signaling pathway in regulating monocyte subsets remains elusive. Here, we demonstrated that RBP-J controlled homeostasis of blood Ly6C^{lo} monocytes in a cell intrinsic manner. RBP-J deficiency led to a drastic increase in blood Ly6C^{lo} monocytes, lung Ly6C^{lo} monocytes and CD16.2^{+} IM. Our results suggest that RBP-J plays a critical role in regulating the population and characteristics of Ly6C^{lo} monocytes in the blood and lung.

Results

RBP-J is essential for maintenance of blood Ly6C^{lo} monocytes

To investigate the role of RBP-J in monocyte subsets, we utilized mice with RBP-J specific deletion in the myeloid cells (*Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice). Efficient deletion of *Rbpj* in blood monocytes was confirmed by quantitative real-time PCR (qPCR) (Figure 1 - figure supplement 1A). Flow cytometric analysis revealed that RBP-J deficient mice had a significant increase in the proportion of Ly6C^{lo} monocytes, but not Ly6C^{hi} monocytes, in blood, compared to age-matched control mice with the genotype *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ (Figure 1A). In contrast to circulating monocytes, RBP-J deficient mice exhibited minimal alterations in the percentages of monocyte subsets in bone marrow (BM) and spleen (Figure 1B; Figure 1 - figure supplement 1C). Additionally, RBP-J deficiency in myeloid cells did not appear to have an effect on neutrophils (Figure 1 - figure supplement 1B). Therefore, among circulating myeloid populations, RBP-J selectively controlled the subset of Ly6C^{lo} monocytes.

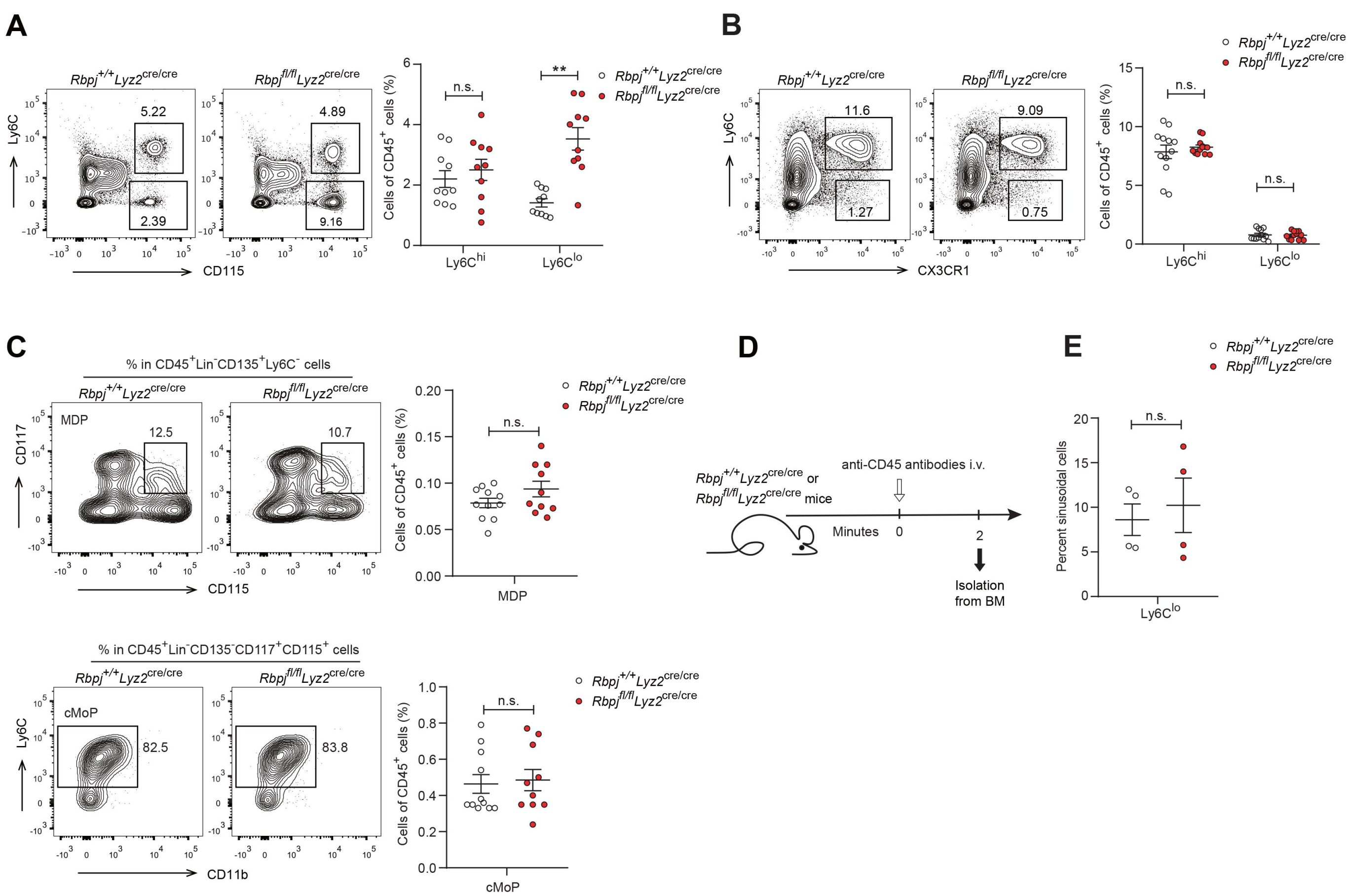

Figure 1.

RBP-J deficient mice display more Ly6C^{lo} blood monocytes.

(A) Blood Ly6C^{hi} and Ly6C^{lo} monocytes in *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice were determined by flow cytometry analyses (FACS). Representative FACS plots (left) and cumulative data of cell ratio (right) are shown.

(B and C) Representative FACS plots and cumulative data quantitating percentages of BM monocyte subsets (B) and myeloid progenitor cells (C) ($CD45^+CD11b^+Ly6G^-$ $CD115^+$ $Ly6C^{hi}$ monocyte; $CD45^+CD11b^+Ly6G^-CD115^+Ly6C^{lo}$ monocyte; MDP, $CD45^+Lin^-CD117^+CD115^+CD135^+$ $Ly6C^-$; cMoP, $CD45^+Lin^-CD11b^-CD117^+CD115^+$ $CD135^-Ly6C^+$). Lin: CD3, B220, Ter119, and Gr-1, CD11b.

(D) Experimental outline for panel (E).

(E) Cumulative data quantitating percentages of sinusoidal monocytes ($CD45^+$) within total BM $Ly6C^{lo}$ monocytes.

Data are pooled from at least 2 independent experiments; n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant; **, P < 0.01 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

BM progenitors that give rise to circulating monocytes are monocyte-dendritic cell progenitors (MDPs) and common monocyte progenitors (cMoPs)[(21),(31)-(33)]. Next, we analyzed BM progenitors, and found that the percentages of MDPs and cMoPs were equivalent in control and RBP-J deficient mice (Figure 1C). These results in conjunction with the observations of normal BM monocyte populations suggest that alterations of peripheral blood $Ly6C^{lo}$ monocytes may not originate from BM. Next, we examined whether RBP-J may influence the egress of $Ly6C^{lo}$ monocytes from BM, and measured BM exit rate of $Ly6C^{lo}$ monocytes using *in vivo* labeling of sinusoidal cells as previously described (Figure 1D)[(34)]. The results showed that RBP-J deficiency did not affect egress of $Ly6C^{lo}$ monocytes from BM (Figure 1E). Taken together, these data revealed RBP-J as a critical regulator controlling homeostasis of peripheral blood $Ly6C^{lo}$ monocytes.

RBP-J deficiency does not affect $Ly6C^{hi}$ monocyte conversion or $Ly6C^{lo}$ monocyte survival and proliferation

Next, we aimed to determine whether the increase of blood $Ly6C^{lo}$ monocytes in RBP-J deficient mice was due to decreased cell death or enhanced proliferation. We first stained monocytes with Annexin V and 7-AAD to identify apoptotic cells and observed comparable percentages of apoptotic blood $Ly6C^{lo}$ monocytes in control and RBP-J deficient mice (Figure 2A). Given the crucial role of Nr4a1 in the survival of $Ly6C^{lo}$ monocytes[(21)], we detected the expression of Nr4a1, which was similar in $Ly6C^{lo}$ monocytes from control and RBP-J deficient mice (Figure 2 - figure supplement 1A). We then assessed the proliferative capacity by analyzing Ki-67 expression as well as *in vivo* EdU incorporation. Ki-67 levels in blood monocytes displayed no differences between control and RBP-J deficient mice (Figure 2B). The percentage of EdU^+ monocytes did not significantly differ between control and RBP-J deficient mice at the indicated time points, implying that the turnover of $Ly6C^{lo}$ monocytes was normal in RBP-J deficient mice (Figure 2C). Fluorescent latex beads as particulate tracers could be phagocytosed by monocytes after intravenous injection, and stably label $Ly6C^{lo}$ monocytes[(35)]. Thus, we intravenously injected fluorescent latex beads into control and RBP-J deficient mice to track circulating monocytes (Figure 2D). By day 7 post injection, only $Ly6C^{lo}$ monocytes were $latex^+$ as previously reported[(35)], whereas control and RBP-J deficient mice groups presented a similar frequency of $latex^+$ monocytes (Figure 2E). Together, these data indicated that RBP-J did not influence monocyte survival and proliferation.

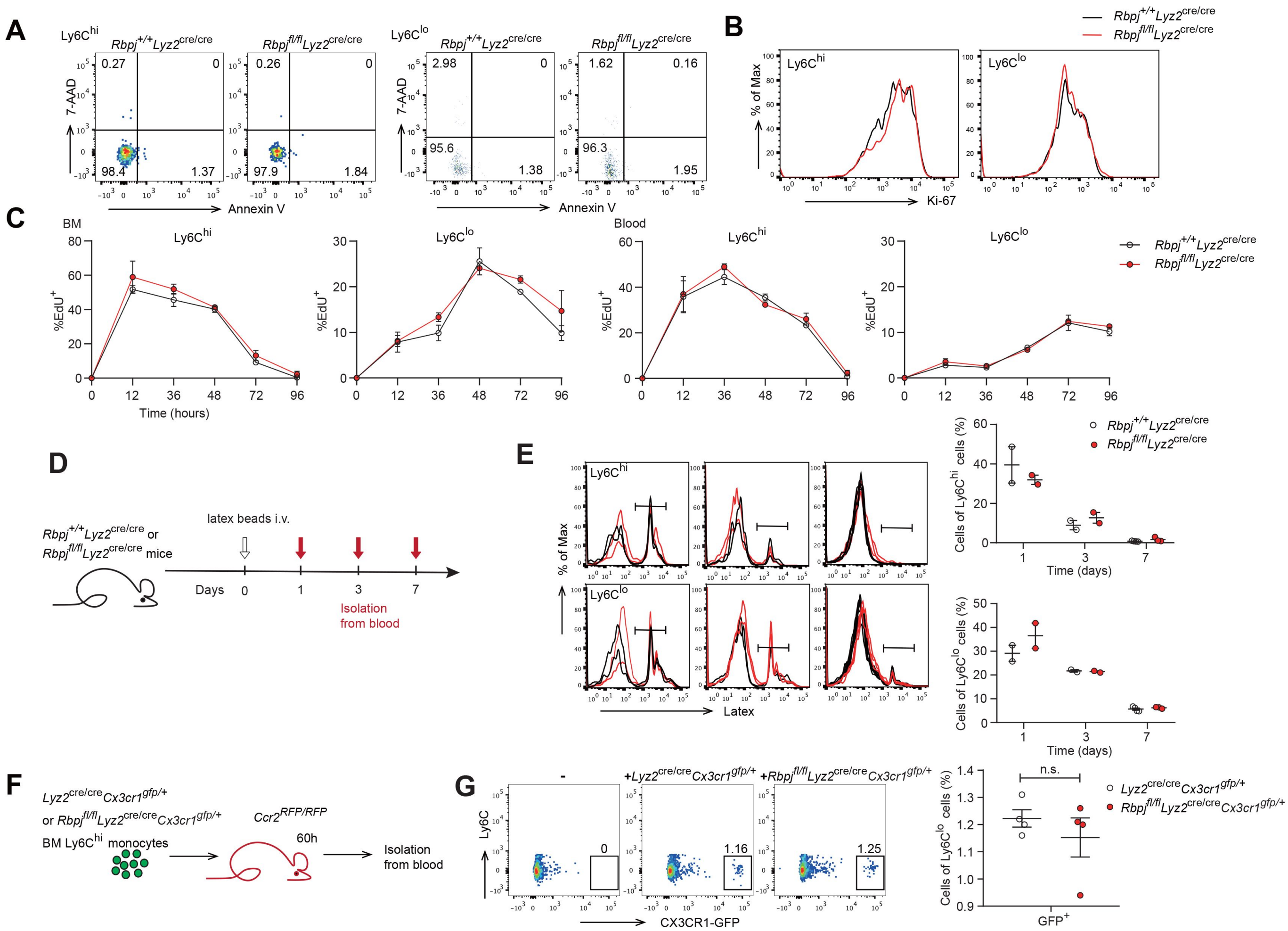

Figure 2.

Monocyte subsets in RBP-J deficient mice display normal cell death.

(A) Representative FACS plots of monocyte subsets in blood stained with 7-AAD and Annexin V.

(B) FACS analysis of Ki-67 expression in *Rbpj*$^{+/+}$*Lyz2*$^{\text{cre/cre}}$ control and *Rbpj*$^{fl/fl}$*Lyz2*$^{\text{cre/cre}}$blood monocyte subsets. Black lines represent control mice and red lines represent RBP-J deficient mice.

(C) Analysis of time course of EdU incorporation of monocyte subsets in BM and blood after a single 1mg EdU pulsing. The percentages of EdU^{+} cells among the indicated monocyte subsets are shown.

(D) Experimental outline for panel (E).

(E) Analysis of time course of latex beads incorporation of monocyte subsets in blood after Latex beads injection. The percentages of latex^{+} cells among the indicated monocyte subsets are shown.

(F) Cartoon depicting the adoptive transfer. BM GFP^{+}Ly6C$^{\text{hi}}$ monocytes were sorted from *Lyz2*$^{\text{cre/cre}}$*Cx3cr1*$^{gfp/+}$or *Rbpj*$^{fl/fl}$*Lyz2*$^{\text{cre/cre}}$*Cx3cr1*$^{gfp/+}$mice and transferred into *Ccr2*$^{RFP/RFP}$ recipient mice. Sixty hours after transfer, cell fate was analyzed.

(G) Representative FACS plots are shown in the left panel, and the frequencies of GFP^{+} Ly6C$^{\text{lo}}$ monocytes within total Ly6C$^{\text{lo}}$ monocytes are shown in the right panel.

Data are pooled from 2 independent experiments (G); n ≥ 2 in each group (C, E and G). Data are shown as mean ± SEM; n.s., not significant; (two-tailed Student's unpaired t test). Each symbol represents an individual mouse (E and G).

Previous studies have shown that Ly6C^{lo} monocytes are observed in recipient mice following the adoptive transfer of Ly6C^{hi} monocytes, indicating that Ly6C^{hi} monocytes can convert into Ly6C^{lo} monocytes[(10),(31)]. We next wished to evaluate whether conversion of Ly6C^{hi} monocytes was regulated by RBP-J by isolating bone marrow GFP^{+}Ly6C^{hi} monocytes from *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$control or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$mice and adoptively transferring them into *Ccr2*$^{RFP/RFP}$ recipients (Figure 2F). Sixty hours after transfer, a subset of cells from donors were converted into Ly6C^{lo} monocytes (Figure 2 - figure supplement 1B), and equal percentages of Ly6C^{lo} monocytes were derived from control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$donors (Figure 2G). Thus, the conversion of Ly6C^{hi} monocytes into Ly6C^{lo} monocytes was not affected by RBP-J deficiency.

RBP-J regulates blood Ly6C^{lo} monocytes in a cell intrinsic manner

We next wondered whether the increase of Ly6C^{lo} monocytes in RBP-J deficient mice was bone marrow-derived and cell intrinsic. We performed bone marrow transplantation by engrafting lethally irradiated mice with a 1:4 mixture of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ BM cells (CD45.2) and *Cx3cr1*$^{gfp/+}$ BM cells (CD45.1) (Figure 3A). Eight weeks after transplantation, we analyzed the frequencies of CD45.2^{+} donor cells in the bone marrow and blood of recipient mice and found that the frequencies of CD45.2^{+} cells within total cells were similar to the mixture ratio of bone marrow cells for Ly6C^{hi} monocytes and neutrophils in both BM and blood (Figure 3B; Figure 3 - figure supplement 1A-B). Specifically, more blood Ly6C^{lo} monocytes were derived from RBP-J deficient donors than control donors (Figure 3B), reflecting that the contribution of RBP-J deficient cells in blood Ly6C^{lo} cells was significantly higher than that of control cells. These results implied that RBP-J deficient BM cells were highly efficient in generating blood Ly6C^{lo} monocytes.

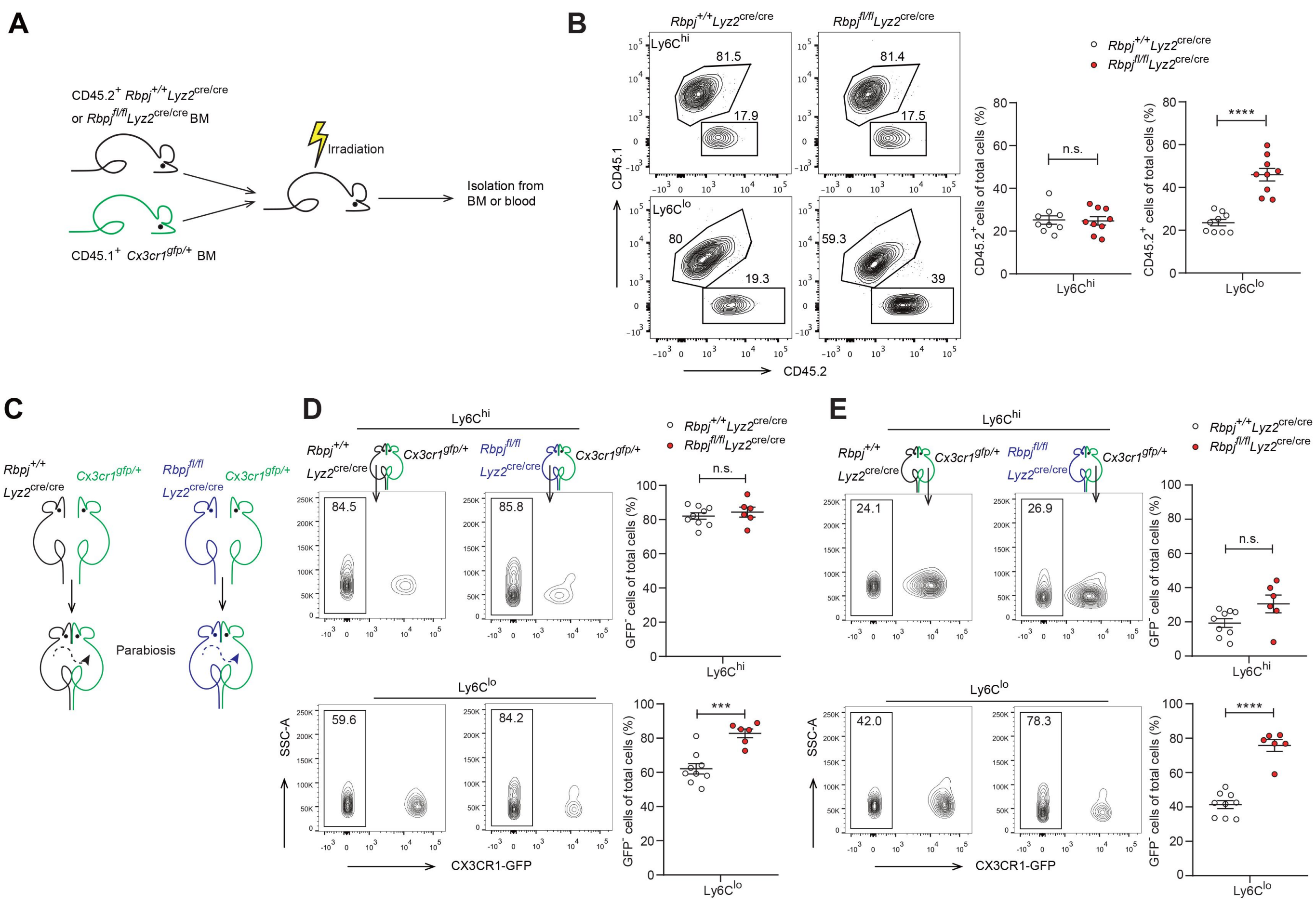

Figure 3.

The role of RBP-J in blood Ly6C^{lo} monocytes is cell intrinsic.

(A) Cartoon depicting the bone marrow transplantation. BM recipient 6-week-old male C57BL/6 mice (CD45.2) were lethally irradiated. BM cells from donor mice (*Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ and CD45.1, *Cx3cr1*$^{gfp/+}$) were collected and transferred into recipient mice. Mice were used after 8 weeks of bone marrow reconstitution.

(B) Representative FACS plots (left) and cumulative data (right) quantitating the frequency of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ donor cells among Ly6C^{hi} and Ly6C^{lo} monocytes in the blood of recipient mice.

(C) Cartoon depicting the generation of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ and *Cx3cr1*$^{gfp/+}$ parabiotic pairs.

(D) Representative FACS plots (left) and cumulative data (right) quantitating percentages of monocyte subsets derived from control or RBP-J deficient mice in control or RBP-J deficient mice.

(E) Representative FACS plots (left) and cumulative data (right) quantitating percentages of monocyte subsets derived from control or RBP-J deficient mice in *Cx3cr1*$^{gfp/+}$ mice.

Data are pooled from at least 2 independent experiments; n ≥ 6 in each group. Data are shown as mean ± SEM; n.s., not significant; ***, P < 0.001; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse. SSC-A, side scatter area.

Given that RBP-J deficient mice had more Ly6C^{lo} monocytes in blood than did control mice, we performed parabiosis experiments, a surgical union of two organisms, which allowed parabiotic mice to share their blood circulation. The contribution of circulating cells from one animal in another can be estimated by measuring the percentage of blood cells that originated from each animal[(36)]. We joined a CD45.1, *Cx3cr1*$^{gfp/+}$ mouse with an age and sex-matched *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control or *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mouse (CD45.2) (Figure 3C). Four weeks after the procedure, about 50% of B cells and T cells in parabiotic mice displayed efficient exchange of their circulation (Figure 3 - figure supplement 1C). As expected, RBP-J deficient mice exhibited higher percentages of Ly6C^{lo} monocytes than control animals (Figure 3D). Intriguingly, in the parabiotic *Cx3cr1*$^{gfp/+}$mice, RBP-J deficient cells still constituted significantly higher proportion of circulating Ly6C^{lo}, but not Ly6C^{hi} monocytes, than RBP-J sufficient counterparts as indicated by the percentages of the GFP-negative population (Figure 3E). These results implicated that the enhanced ability of Ly6C^{lo} monocytes to circulate in the peripheral blood as a result of RBP-J deficiency was cell intrinsic.

RBP-J regulates phenotypical marker genes in blood Ly6C^{lo} monocytes

To further study the consequences of RBP-J loss of function in blood Ly6C^{lo} monocytes, we performed gene expression profiling by RNA-seq, which revealed that Ly6C^{lo} monocytes in RBP-J deficient mice exhibited low expression of *Itgax* but high expression of *Ccr2*, in comparison to those in control mice (Figure 4A). These differential expression patterns of phenotypic marker genes were also confirmed by qPCR and flow cytometry analysis (Figure 4B). However, Ly6C^{hi} blood monocytes and BM monocytes displayed normal levels of CD11c and CCR2 (Figure 4B; Figure 4 - figure supplement 1A). Moreover, principal component analysis showed distinct expression pattern of monocytes between control and RBP-J deficient mice (Figure 4C). To determine whether the regulation of phenotypic markers by RBP-J in blood Ly6C^{lo} monocytes was cell-intrinsic, we performed BM transplantation as described above (Figure 3A). The findings indicated that Ly6C^{lo} monocytes derived from RBP-J deficient BM cells expressed high levels of CCR2 but low levels of CD11c, in comparison to those derived from control BM cells, whereas no significant changes were observed in blood Ly6C^{hi} and BM monocytes, as expected (Figure 4D; Figure 4 - figure supplement 1B). These results collectively suggested that RBP-J regulated the expression of CCR2/CD11c in a cell-intrinsic manner.

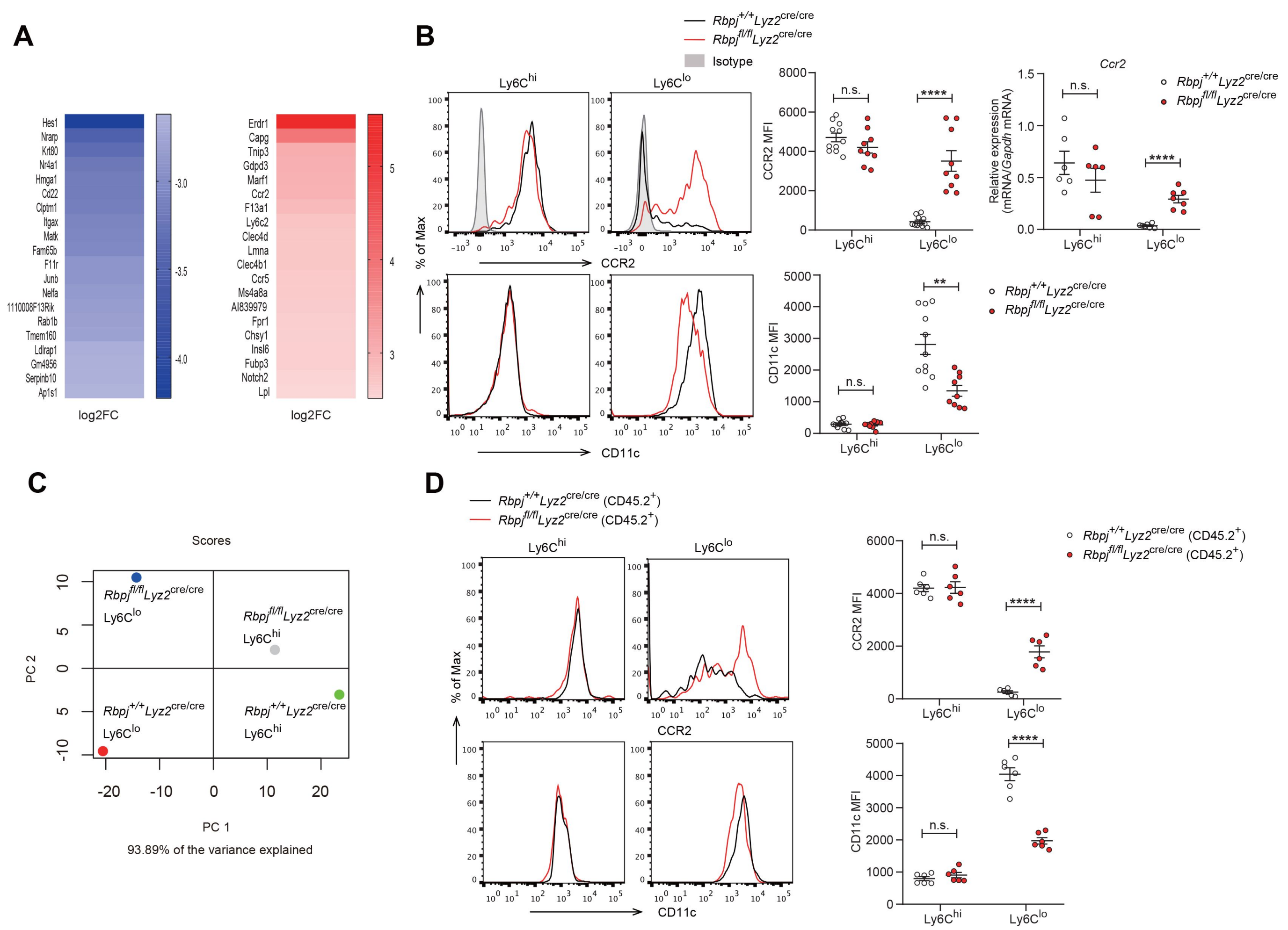

Figure 4.

Phenotypic markers in Ly6C^{lo} monocytes are changed in RBP-J deficient mice.

(A) Heatmap of RNA-seq dataset showing the top 20 downregulated and upregulated genes in blood Ly6C^{lo} monocytes from *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ versus *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ control mice. Blue and red font indicates downregulated and upregulated genes in RBP-J deficient Ly6C^{lo} monocytes respectively.

(B) Representative FACS plots, cumulative MFI, and qPCR analysis of CCR2/CD11c expression in control and RBP-J deficient blood monocyte subsets. Shaded curves represent isotype control, black lines represent control mice and red lines represent RBP-J deficient mice.

(C) PCA of indicated cell types.

(D) Representative FACS plots and cumulative MFI of CCR2/CD11c expression in blood monocyte subsets derived from control or RBP-J deficient mice. Black lines represent control mice, and red lines represent RBP-J deficient mice.

Data are pooled from 2 independent experiments (B and D); n ≥ 6 in each group. Data are shown as mean ± SEM; n.s., not significant; **, P < 0.01; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

RBP-J-mediated control of blood Ly6C^{lo} monocytes is CCR2 dependent

Given that RBP-J deficiency led to enhanced CCR2 expression in Ly6C^{lo} monocytes and that CCR2 is essential for monocyte functionality under various inflammatory and non-inflammatory conditions[37], we wished to genetically investigate the role of CCR2 in the RBP-J deficient background. We generated RBP-J/CCR2 double deficient (DKO) mice with the genotype *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ with *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$ mice serving as controls. As expected, the expression of CCR2 in Ly6C^{lo} monocytes was reduced in DKO mice compared to RBP-J deficient mice (Figure 5A), confirming the successful deletion of the *Ccr2* gene. DKO mice showed a lower percentage of both Ly6C^{hi} and Ly6C^{lo} monocytes than RBP-J deficient mice (Figure 5B-C). Notably, the percentage of Ly6C^{lo} monocytes in DKO mice was comparable to that observed in the control mice (Figure 5B-C), implicating that deletion of CCR2 corrected RBP-J deficiency-associated phenotype of increased Ly6C^{lo} monocytes. Whereas, both RBP-J deficient and DKO mice exhibited lower expression levels of CD11c in their Ly6C^{lo} monocyte than control mice, suggesting that the regulation of CD11c expression was independent of CCR2 (Figure 5D). These results suggested that RBP-J regulated Ly6C^{lo} monocytes, at least in part, through CCR2.

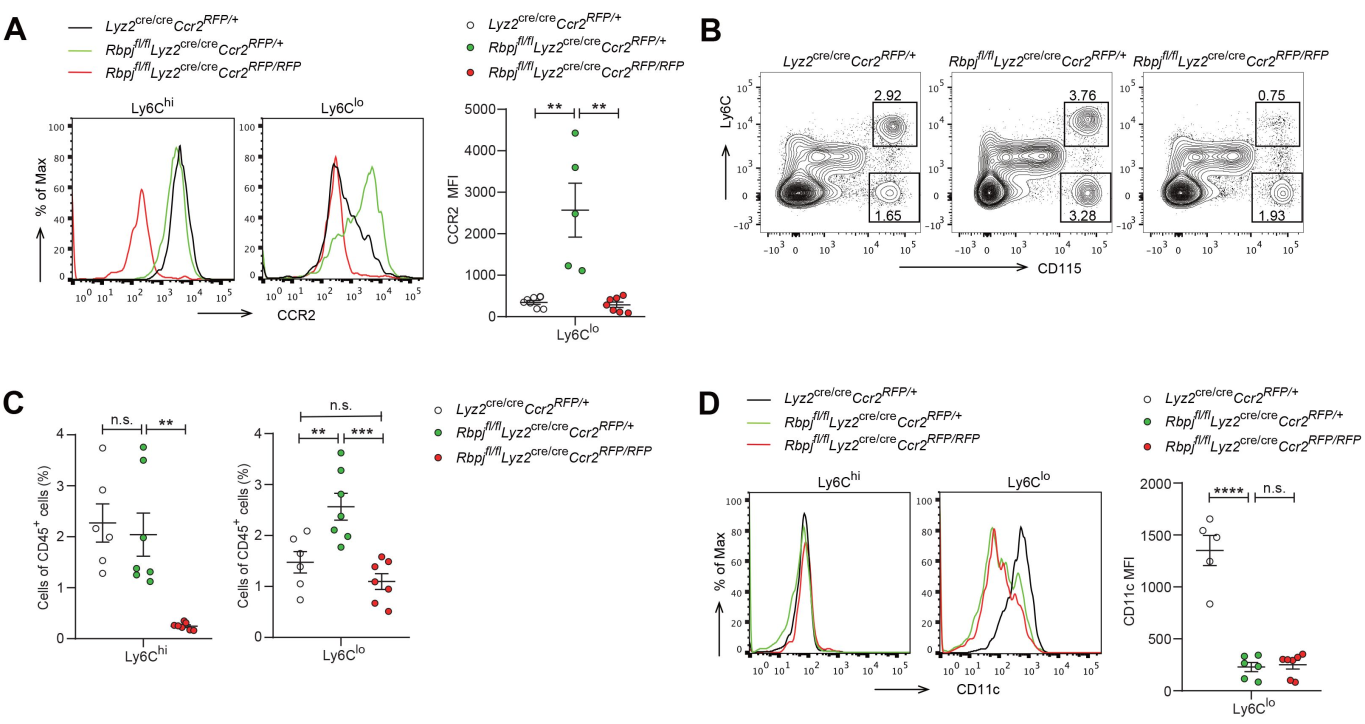

Figure 5.

Blood Ly6C^{lo} monocytes are decreased in DKO mice.

(A) Representative FACS plots and cumulative MFI of CCR2 expression in *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$ control, *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ (DKO) blood Ly6C^{hi} and Ly6C^{lo} monocytes are shown.

(B and C) Blood monocyte subsets in control, RBP-J deficient and DKO mice were determined by FACS. Representative FACS plots (B) and cumulative data of cell ratio

(C) are shown.

(D) Representative FACS plots and cumulative MFI of CD11c expression in control, RBP-J deficient and DKO blood Ly6C^{hi} and Ly6C^{lo} monocytes are shown.

Data are pooled from at least 2 independent experiments; n ≥ 5 in each group. Data are shown as mean ± SEM; n.s., not significant; **, P < 0.01; ***, P < 0.001; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

Lung Ly6C^{lo} monocytes are accumulated in RBP-J deficient mice

In mice, alveolar macrophages (AM) are maintained by local self-renewal, and arise from fetal liver derived precursors, whereas interstitial macrophages (IM) are probably originated from monocytes [(38)-(40)]. Joey et al identified 3 subpopulations of IM according to expression of CD206 and CD16.2 and Ly6C^{lo} monocytes are proposed to give rise to CD64^{+}CD16.2^{+} IM[(41)]. We next compared the populations of lung monocytes and IM between control and RBP-J deficient mice. The conditional deletion of RBP-J resulted in a significant increase in the absolute and relative numbers of Ly6C^{lo} monocytes and CD16.2^{+} IM, while the numbers of Ly6C^{hi} monocytes, CD206^{-} IM, and CD206^{+} IM remained unchanged (Figure 6A-B). To confirm these findings, lung sections from *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ control and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ mice were stained with an anti-GFP antibody, which revealed a significant increase in the number of GFP^{+} cells in RBP-J deficient mice (Figure 6C). In addition, we evaluated the proliferative capacity of Ly6C^{lo} monocytes and CD16.2^{+} IM, but did not observe enhanced EdU incorporation in Ly6C^{lo} monocytes and CD16.2^{+} IM from RBP-J deficient mice (Figure 6 - figure supplement 1A-B). These findings suggested that the augmented population of these cells might not be resulted from increased *in situ* proliferation. Previous reports have indicated that LPS can increase the population of IM in a CCR2-dependent manner[(40)]. We thus challenged mice with LPS, and analyzed monocytes and IM at day 0 and day 4. LPS exposure induced an increase in numbers of Ly6C^{lo} monocytes, CD16.2^{+} IM and CD206^{-} IM compared to baseline both in control and RBP-J deficient mice (Figure 6D-E). There was a trend toward higher Ly6C^{hi} monocyte after LPS treatment, although this observation was not statistically significant. Of note, RBP-J deficient mice exhibited robustly elevated numbers of Ly6C^{lo} monocytes and CD16.2^{+} IM compared with control mice after LPS treatment (Figure 6D-E), suggesting that RBP-J deficient Ly6C^{lo} monocytes were recruited to inflamed tissue in large numbers and differentiated into CD16.2^{+} IM.

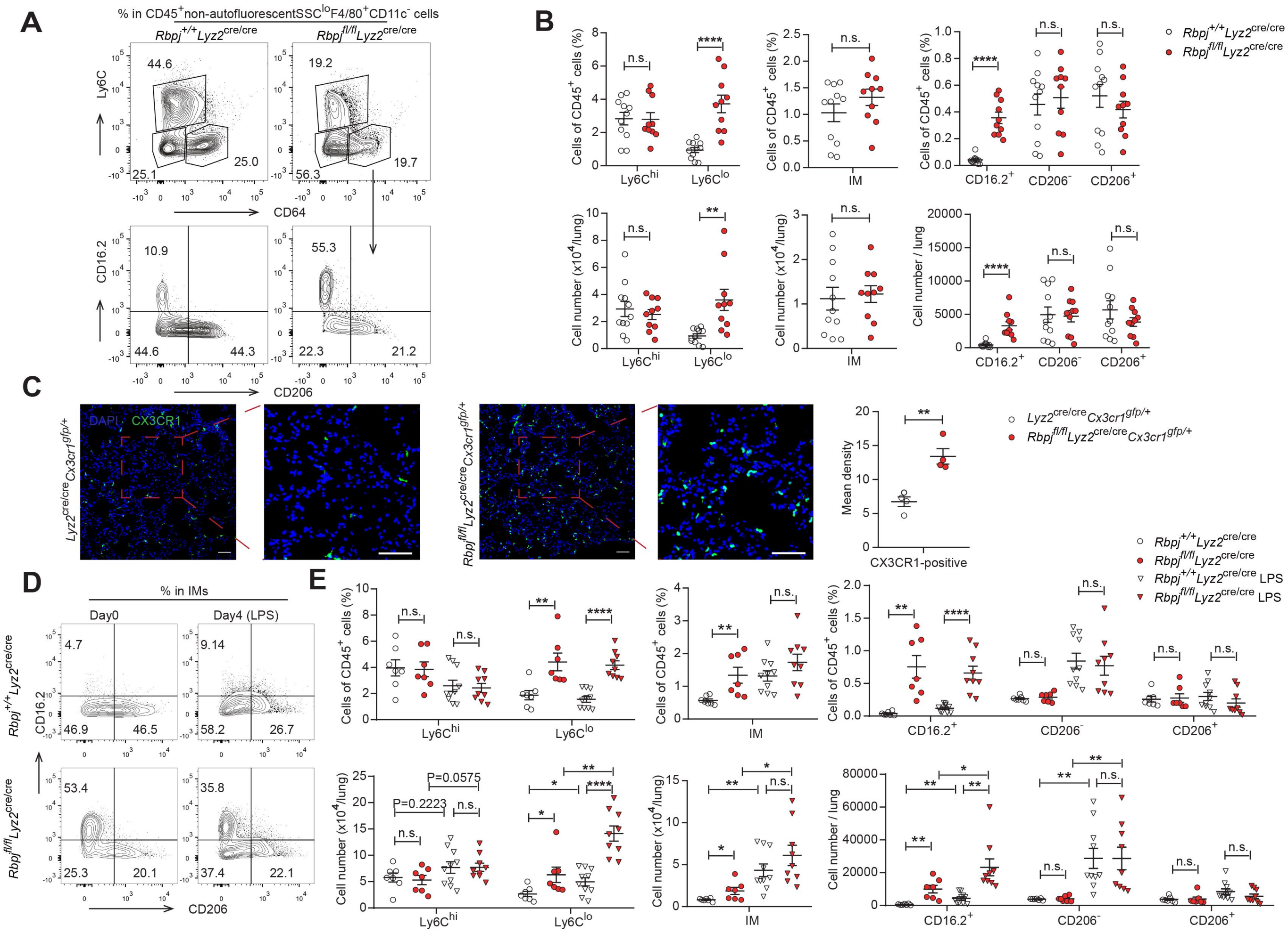

Figure 6.

RBP-J deficient mice exhibit more lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

(A and B) Indicated populations in the lungs of *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice were determined by FACS. Representative FACS plots (A) and cumulative data of cell ratio and absolute numbers (B) are shown.

(C) Immunofluorescence staining for GFP^{+} cells in the lung from *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$mice (CX3CR1 [green]; DAPI [blue]). Scale bars represent 50 µm.

(D and E) *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice were instilled intranasally with PBS or PBS containing LPS, and lungs were harvested at the indicated time points. Representative FACS plots (D) and cumulative data of cell ratio and absolute numbers

(E) are shown.

Data are pooled from at least 2 independent experiments; n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant; *, P < 0.05; **, P < 0.01; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

To investigate whether the elevated levels of lung monocytes and CD16.2^{+} IM in RBP-J deficient mice were derived from blood Ly6C^{lo} monocytes, we examined monocytes and IM in *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$control, *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$, *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$(DKO) mice. The deletion of CCR2 led to a decrease in Ly6C^{hi} monocytes, and the Ly6C^{lo} monocytes in DKO mice were reduced to a level similar to that of control mice (Figure 7A, B). While the DKO mice had more CD16.2^{+} IM compared to control mice, these cells were severely reduced compared to RBP-J deficient mice (Figure 7A, C). The above data supported the notion that increased lung Ly6C^{lo} monocytes and CD16.2^{+} IM in RBP-J deficient mice were derived from Ly6C^{lo} blood monocytes. In summary, these results suggested that in RBP-J deficient mice, recruitment of blood Ly6C^{lo} monocytes to the lung was markedly facilitated by increased cell number as well as heightened expression of CCR2, leading to the increase of lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

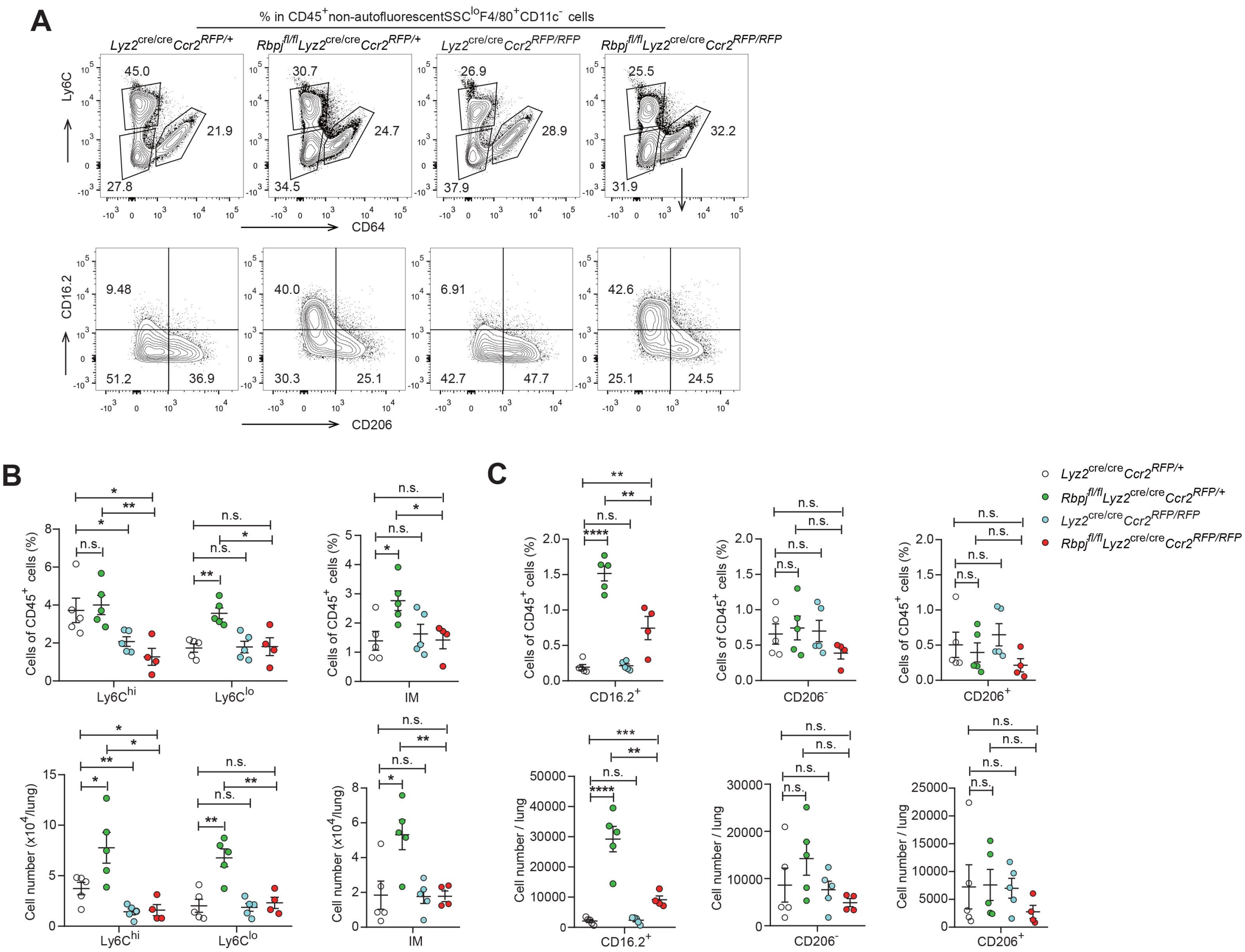

Figure 7.

DKO mice lack lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

(A) Representative FACS plots of lung monocyte and IM subsets in *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ mice.

(B and C) Cumulative data of cell ratio and absolute numbers of monocyte (B) and IM (C) subsets.

Data are pooled from 2 independent experiments; n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant; *, P < 0.05; **, P < 0.01; ***, P < 0.001; ****, P < 0.0001 (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

Discussion

In this study, we identified RBP-J as a pivotal factor in regulating blood Ly6C^{lo} monocyte cell fate. Our results showed that mice with conditional deletion of RBP-J in myeloid cells exhibited a robust increase in blood Ly6C^{lo} monocytes, and subsequently accumulated lung Ly6C^{lo} monocytes and CD16.2^{+} IM under steady-state conditions. Bone marrow transplantation experiments in which RBP-J deficient cells were transplanted into recipient mice as well as the parabiosis experiment showed similar increase in blood Ly6C^{lo} monocytes, demonstrating that RBP-J was an intrinsic factor required for the regulation of Ly6C^{lo} monocytes. Further analysis revealed that RBP-J regulated the expression of CCR2 and CD11c on the surface of Ly6C^{lo} monocytes. Moreover, the phenotype of elevated blood Ly6C^{lo} monocytes and their progeny was further ameliorated by deleting *Ccr2* in an RBP-J deficient background.

Notch-RBP-J signaling have been shown to play a role in regulating functional polarization and activation of macrophages, and regulated formation of Kupffer cells and macrophage differentiation from Ly6C^{hi} monocytes in ischemia(42)-(48). At steady-state, interaction of DLL1 with Notch 2 regulates conversion of Ly6C^{hi} monocytes into Ly6C^{lo} monocytes in special niches of the BM and spleen(49). Under inflammatory conditions, Notch2 and TLR7 pathways independently and synergistically promote conversion of Ly6C^{hi} monocytes into Ly6C^{lo} monocytes(50). However, in our study, the percentage of blood Ly6C^{lo} monocytes increased in RBP-J deficient mice. These results somewhat differ from what was observed in mice with conditional deletion of Notch2(49). Actually, 4 members of the Notch family have been identified in mammals(26). Thus, Notch receptors seem to be non-redundant in regulating monocyte cell fate, and distinct Notch receptors are likely to act in a concerted manner to coordinate the monocyte differentiation program.

Colony-stimulating factor 1 receptor (CSF1R) signal, CX3CR1, C/EBPβ and Nr4a1 have been suggested to be involved in the generation and survival of Ly6C^{lo} monocytes. The lifespan of Ly6C^{lo} monocytes is acutely shortened after blockade of CSF1R, and the numbers of these cells are reduced, and the percentage of dead cells increased in mice with endothelial cell-specific depletion of Csf1(23),(24). CX3CR1 and C/EBPβ-knockout mice show accelerated death of Ly6C^{lo} monocyte, display the decreased level of Ly6C^{lo} monocytes(22),(51). Nr4a1 knockout mice had significantly fewer monocytes in blood, bone marrow, and spleen, due to differentiation deficiency in MDP and accelerated apoptosis of Ly6C^{lo} monocytes in bone marrow(21). RBP-J deficient mice had normal MDP and Ly6C^{lo} monocytes, expressed normal level of Nr4a1 and Ki-67, exhibited normal half-life and turnover rate, which suggested that RBP-J may not regulate Ly6C^{lo} monocytes through these factors. Further studies are required to elucidate the precise molecular mechanisms of RBP-J in Ly6C^{lo} monocytes under steady state.

At the steady state, Ly6C^{lo} monocytes patrol endothelium of blood vessel. After R848 induced endothelial injury, Ly6C^{lo} monocytes recruit neutrophils to mediate focal endothelial necrosis and fuel vascular inflammation, and subsequently, the Ly6C^{lo} monocytes remove cellular debris(9),(52). In response to *Listeria monocytogenes* infection, Ly6C^{lo} monocytes can unmethylated CpG DNA expands $CD16.2^+$ IM, which originate from $Ly6C^{lo}$ monocytes and spontaneously produce IL-10, thereby preventing allergic inflammation[41]. A recent study shows that $Ly6C^{lo}$ monocytes give rise to $CD9^+$ macrophages, which provide an intracellular replication niche for *Salmonella* Typhimurium in spleen, and $Ly6C^{lo}$-depleted mice are more resistant to *Salmonella* Typhimurium infection, suggesting that $Ly6C^{lo}$ monocytes exert certain functions in systemic infection[53]. Our data provide evidence for RBP-J in the function of blood $Ly6C^{lo}$ monocytes, as RBP-J deficient $Ly6C^{lo}$ monocytes exhibited enhanced competition in blood circulation, as well as gave rise to increased numbers of lung $CD16.2^+$ IM. In summary, we showed that RBP-J acted as a fundamental regulator of the maintenance of blood $Ly6C^{lo}$ monocytes and their descendent, at least in part through regulation of CCR2. These results provide insights into understanding the mechanisms that regulate monocyte homeostasis and function.

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Author contributions

T. Kou designed and performed experiments, analyzed and interpreted data, and wrote the manuscript. L. Kang designed and performed experiments, analyzed and interpreted data. B. Zhang provided advice on RNA-seq analyses. J. Li performed experiments. B. Zhao provided advice on experiments. W. Zeng provided advice on experiments. X. Hu conceptualized the project, supervised experiments, interpreted data, and wrote the manuscript.

Declaration of interests

The authors declare no competing interests.

Data availability

Sequencing data sets are deposited in the Genome Expression Omnibus with assigned accession numbers as follows: RNA-seq (GEO accession no. GSE208772).

Materials and methods

Mice

Cx3cr1$^{gfp/gfp}$mice (JAX stock 005582; Jung et al., 2000) and *Ccr2*$^{RFP/RFP}$ mice (JAX stock 017586) were purchased from the Jackson Laboratory. Mice with a myeloid-specific deletion of the *Rbpj* were generated by crossing *Rbpj*$^{fl/fl}$ mice to *Lyz2*-Cre mice as described previously (Hu et al., 2008). *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ were crossed to *Cx3cr1*$^{gfp/gfp}$mice to obtain *Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Cx3cr1*$^{gfp/+}$ mice. C*x3cr1*$^{gfp/+}$ mice were obtained by crossing *Cx3cr1*$^{gfp/gfp}$with C57/BL6 CD45.1^{+} mice. *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice were crossed with *Ccr2*$^{RFP/RFP}$ mice to obtain *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$, *Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$, *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/+}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$*Ccr2*$^{RFP/RFP}$ mice. All mice were maintained under specific pathogen-free conditions. All animal experimental protocols were approved by the Institutional Animal Care and Use Committees of Tsinghua University. Gender- and age-matched mice were used at 7–12 weeks old for experiments.

Quantitative RT-PCR

Blood monocytes were sorted by flow cytometry. The total RNA was extracted using total RNA purification kit (GeneMarkbio) and reversely transcribed to cDNA by M-MLV Reverse Transcriptase (Takara). qPCR was performed on a real-time PCR system (StepOnePlus; Applied Biosystems) using FastSYBR mixture (CWBIO). *Gapdh* messenger RNA was used as internal control to normalize the expression of target genes. Primer sequences are provided in supplemental Table 1.

Table 1:

Primers sequences for regular qPCR used in this study

Gene	Forward primer	Reverse primer
Gapdh	ATCAAGAAGGTGGTGAAGCA	AGACAACCTGGTCCTCAGTGT
Rbpj	ACCCCTGTGCCTGTCGTAGA A	TCCCGGAATGCAGAAATGTC
Nr4a1	TTGAGCTTGAATACAGGGCA	AGTTGGGGGAGTGTGCTAGA
Ccr2	CCTTGGGAATGAGTAACTGT GTGAT	ATGGAGAGATACCTTCGGAAC TTCT

RNA-seq Analysis

Ly6C^{hi} and Ly6C^{lo} blood monocytes were isolated from *Rbpj*$^{+/+}$*Lyz2*$^{cre/cre}$ and *Rbpj*$^{fl/fl}$*Lyz2*$^{cre/cre}$ mice, and total RNA was extracted using total RNA purification kit (GeneMarkbio). RNA was converted into RNA-seq libraries, which were sequenced with the pair-end option using an Illumina-HiSeq2500 platform at Beijing Genomics Institute (BGI), China. The significantly down-regulated genes were identified with P value <0.05 and (FPKM + 1) fold changes ≤0.2, and significantly up-regulated genes were identified with P value <0.05 and (FPKM + 1) fold changes ≥5.7. The RNA-seq data are deposited in Gene Expression Omnibus under accession number GSE208772

Annexin V staining

Annexin V and 7-amino-actinomycin D (7-AAD) were used for identification of apoptotic monocytes by flow cytometry. Blood cells were stained with annexin V (eBioscience) and 7-AAD (Biolegend) according to the manufacturers' protocols.

EdU pulsing and latex beads labeling

Mice were injected intravenously with a single 1mg EdU (Thermo Scientific). BM and blood cells were collected and stained with Fluorescence-conjugated mAb against CD45, CD11b, Ly6G, CD115 and Ly6C. Cells were then fixed, permeabilized and stained with reaction cocktail using EdU Assay Kit. Labeled cells were analyzed by flow cytometry.

Mice were injected intravenously with a single 10 µL Latex beads (Polysciences) in 250 µl PBS. Blood cells were harvested at indicated time. Cells were then stained for CD45, CD11b, Ly6G, CD115, Ly6C, and analyzed by flow cytometry.

In vivo labeling of sinusoidal leukocytes

Mice were injected intravenously with 1 µg of PE-conjugated anti-CD45 antibodies. Two minutes after antibody injection, mice were sacrificed, and bone marrow were harvested. BM cells were then stained and analyzed by flow cytometry.

Generation of BM chimera

C57/BL6 CD45.2^{+} mice were lethally irradiated in two doses of 5.5Gy 2 h apart. 0.8×10^{5} BM cells from *Rbpj*$^{+/+}$*Lyz2*$^{\mathrm{cre/cre}}$ mice (CD45.2^{+}) or *Rbpj*$^{fl/fl}$*Lyz2*$^{\mathrm{cre/cre}}$ mice (CD45.2^{+}) were mixed with 3.2×10^{5} BM cells from *Cx3cr1*$^{gfp/+}$ mice (CD45.1^{+}), and injected intravenously into recipient mice. Mice were used for experiments 8 weeks after irradiation.

Adoptive transfers

BM GFP^{+}Ly6C$^{\mathrm{hi}}$ monocytes were sorted from *Lyz2*$^{\mathrm{cre/cre}}$*Cx3cr1*$^{gfp/+}$or *Rbpj*$^{fl/fl}$*Lyz2*$^{\mathrm{cre/cre}}$*Cx3cr1*$^{gfp/+}$ mice, and transferred to *Ccr2*$^{RFP/RFP}$ mice. Blood cells were collected 60 hours later for flow cytometry analyses.

Parabiosis

Cx3cr1$^{gfp/+}$CD45.1^{+} mice were surgically joined with age-matched female *Rbpj*$^{fl/fl}$*Lyz2*$^{\mathrm{cre/cre}}$ or *Rbpj*$^{+/+}$*Lyz2*$^{\mathrm{cre/cre}}$ CD45.2^{+} mice at the age of 6 weeks. Bloods were obtained via cardiac puncture, and cell populations were analyzed by flow cytometry at 4 weeks after the surgery.

Intranasal instillations of LPS

Rbpj$^{+/+}$*Lyz2*$^{\mathrm{cre/cre}}$ or *Rbpj*$^{fl/fl}$*Lyz2*$^{\mathrm{cre/cre}}$ mice were anesthetized with isoflurane and intranasally instilled with 10 µg LPS in 25 µl of PBS. Lungs were harvested 4 days later for flow cytometry analyses.

Immunofluorescence histology

Lungs from $Rbpj^{fl/fl}Lyz2^{\mathrm{cre/cre}}Cx3cr1^{gfp/+}$ and $Lyz2^{\mathrm{cre/cre}}Cx3cr1^{gfp/+}$ mice were fixed in 1% paraformaldehyde and incubated in 30% sucrose separately overnight at 4°C. The samples were then incubated in the mixture of 30% sucrose and OTC compound (Sakura Finetek) overnight at 4°C. The tissues were embedded and frozen in OCT compound and then cut at 10-μm thickness. Tissue sections were dried for 10 min at 50°C and then fixed in 1% paraformaldehyde at room temperature for 10 min, permeabilized in PBS/0.5% Triton X-100/0.3 M glycine for 10 min and blocked in PBS/5% goat serum for 1 h at room temperature. Sections were stained with rabbit anti-mouse GFP antibodies (1:1000; EASYBIO) overnight at 4°C, and washed with PBS/0.1% Tween-20 for 30 min three times at room temperature. Sections were then incubated with AF488-conjugated goat anti-rabbit antibodies (1:200; proteintech) for 2 h at room temperature, and washed with PBS/0.1% Tween-20 for 30 min three times at room temperature. Sections were stained with DAPI (Solarbio) for 7 min, washed in PBS for 8 min two times at room temperature and mounted with SlowFade Diamond Antifade Mountant (Life Technologies).

Cell isolation and Flow cytometry

Peripheral blood was sampled by eyeball extirpating, spleens were mashed through a 70-μm strainer, and bone marrow cells were collected from femurs. Lungs were perfused with 5ml of HBSS (MACGENE) through the right ventricle, excised, and digested in HBSS containing 5% FBS, 1 mg/ml collagenase type I (Sigma-Aldrich) and 0.05 mg/ml DNase I (Sigma-Aldrich) for 1 h at 37 °C. The digested tissues were homogenized by shaking, passed through a 70-μm cell strainer to create a single cell suspension. The suspension was enriched in mononuclear cells and harvested from the 1.080:1.038 g ml-1 interface using a density gradient (Percoll from GE Healthcare). BM, Spleen, PB and lung cells were stained with fluorescently conjugated antibodies. The absolute number of cells was counted by using CountBright™ Absolute Counting Beads (Invitrogen).

Antibodies against CD45 (30-F11), Ly6C (HK1.4), CD64 (X54-5/7.1), CD206 (C068C2), CD16.2 (9E9), CD45.1 (A20), F4/80 (BM8), Biotin, CCR2 (SA203G11) and CD3ε (145-2C11) were purchased from BioLegend. Antibodies against Ly6G (1A8-Ly6g), CD11b (M1/70), CD115(AFS98), CD117(2B8), CD135 (A2F10), CD11c (N418), CD206 (MR6F3), CD45.2 (104), Nur77 (12.14), Ki-67 (SolA15) and lineage maker were purchased from eBioscience. Fluorescence-conjugated mAb against Ly6C (AL-21) were purchased from BD Biosciences. Isotype-matched antibodies (eBioscience) were used for control staining. All antibodies were used in 1:400 dilutions, and surface antigens were stained on ice for 30 min.

For intracellular staining, cells were stained with antibodies to surface antigens, fixed and permeabilized with the Cytofix/Cytoperm Fixation/Permeabilization Solution Kit (BD Biosciences), and stained with antibodies or isotype control diluted in permeabilization buffer separately for 30 min at room temperature.

Cells were analyzed on FACSFortessa or FACSAria III flow cytometer (BD Biosciences) using FlowJo software.

Figure Legends

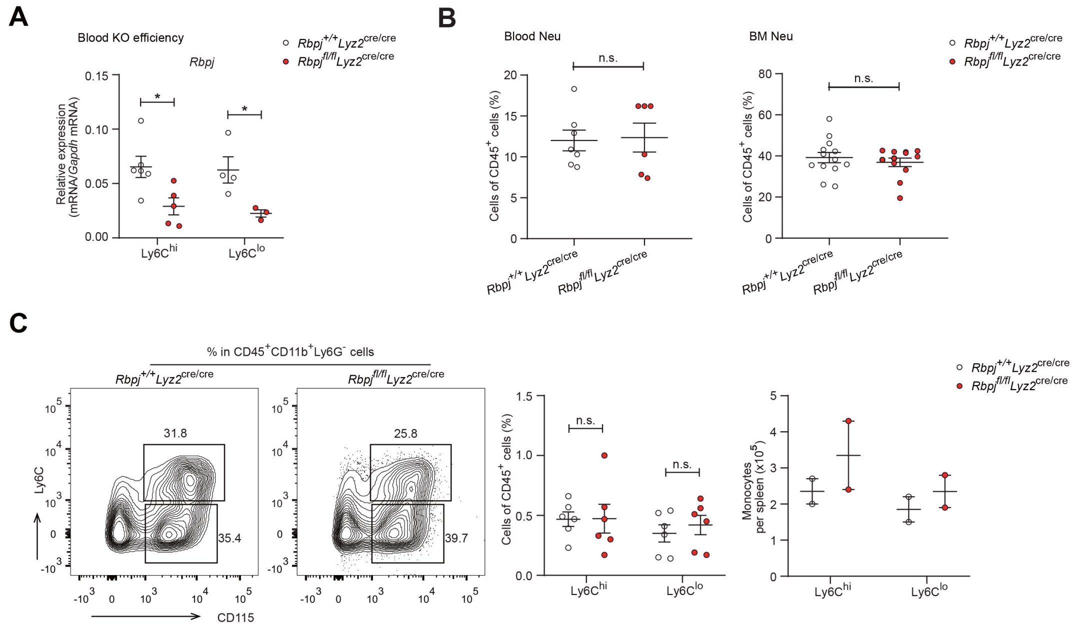


Figure 1 - Figure supplement 1.

Control and RBP-J deficient mice show similar neutrophils.

(A) qPCR analysis of *Rbpj* expression in sorted monocyte subsets from control and RBP-J deficient mice.

(B) Blood and BM $CD45^{+}CD11b^{+}Ly6G^{+}$ neutrophils were determined by FACS. Cumulative data of cell ratio are shown.

(C) Spleen monocyte subsets were determined by FACS. Representative FACS plots (left) and cumulative data of cell ratio and absolute numbers (right) are shown.

Data are pooled from at least 2 independent experiments; n ≥ 2 in each group. Data are shown as mean ± SEM; n.s., not significant; *, $P < 0.05$ (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

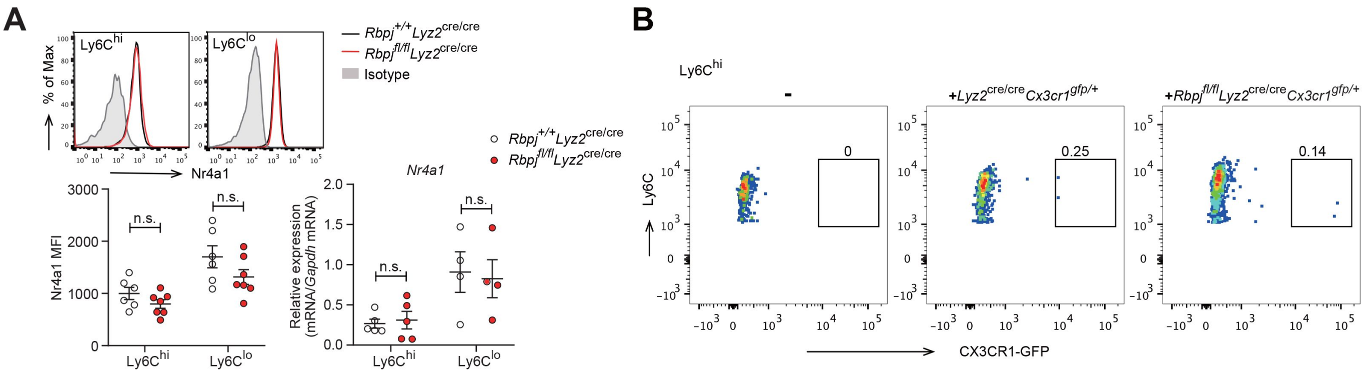


Figure 2 - Figure supplement 1.

The conversion of Ly6C^{hi} monocytes is identical in control and RBP-J deficient mice.

(A) Representative FACS plots, cumulative MFI and qPCR analysis of Nr4a1 expression in control and RBP-J deficient blood monocyte subsets. Shaded curves represent isotype control, black lines represent control mice, and red lines represent RBP-J deficient mice.

(B) Representative FACS plots of GFP^{+} Ly6C^{hi} monocytes in recipient.

Data are pooled from 2 independent experiments (A); n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

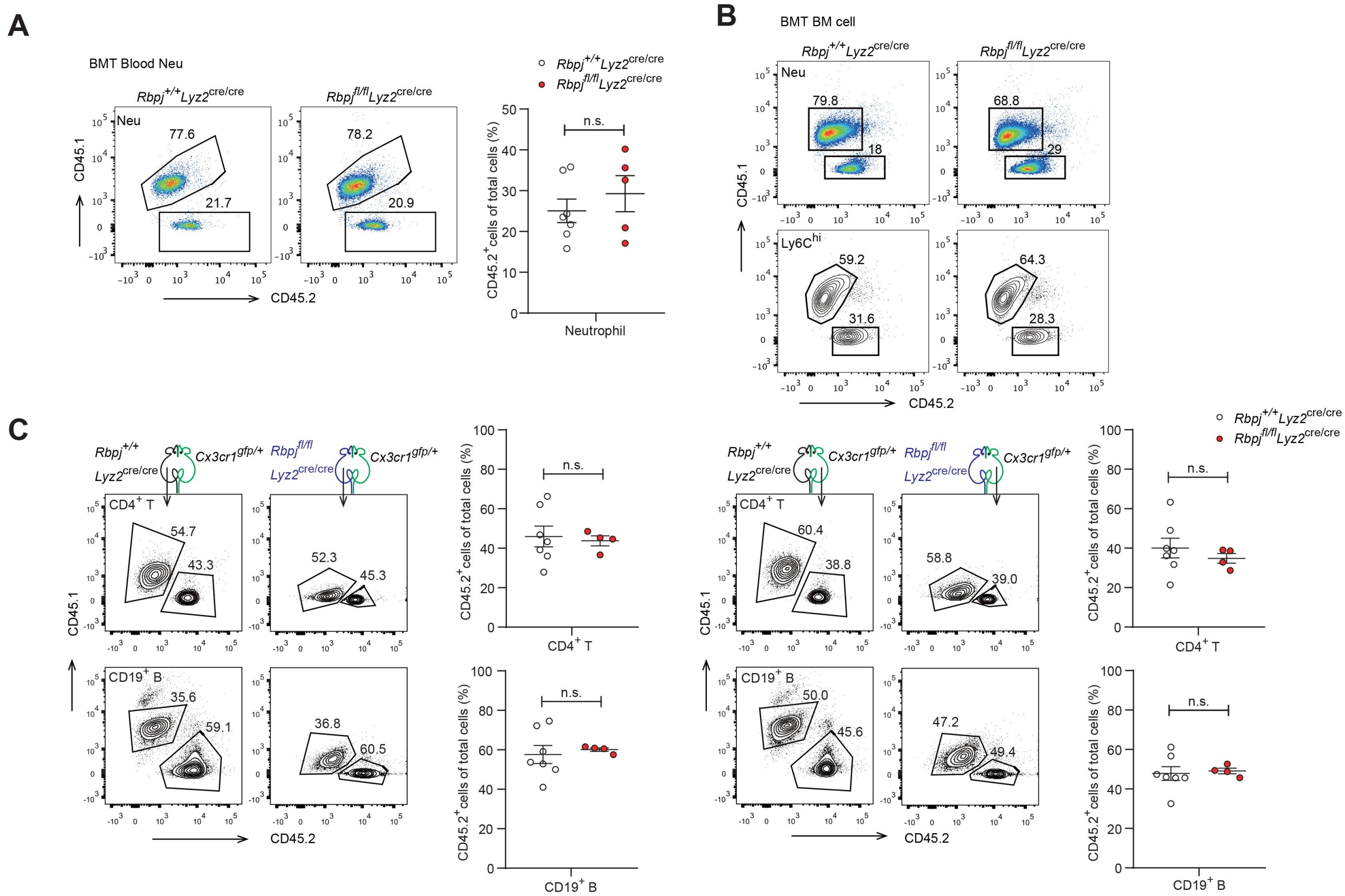


Figure 3 - Figure supplement 1.

Cell-intrinsic requirement of RBP-J for Ly6C^{lo} maintenance.

(A) Representative FACS plots (left) and cumulative data (right) quantitating percentages of blood neutrophils in recipient mice.

(B) BM neutrophils and Ly6C^{hi} monocytes in recipient mice were determined by FACS.

(C) CD4^{+} T cells and CD19^{+} B cells were analyzed by FACS in parabiotic mice.

Data are pooled from at least 2 independent experiments (A and C); n ≥ 4 in each group. Data are shown as mean ± SEM; n.s., not significant (two-tailed Student’s unpaired t test). Each symbol represents an individual mouse.

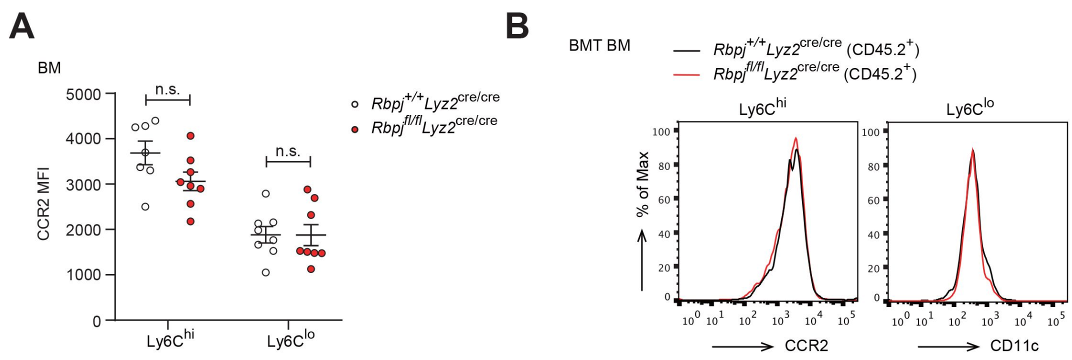


Figure 4 - Figure supplement 1.

Normal expression of CCR2 and CD11c in BM monocytes.

(A) Cumulative MFI of CCR2 expression in BM monocyte subsets.

(B) FACS plots of CCR2 and CD11c in BM Ly6C^{hi} and Ly6C^{lo} from control and RBP-J deficient mice.

Data are pooled from 3 independent experiments (A); n ≥ 7 in each group. Data are shown as mean ± SEM; n.s., not significant (two-tailed Student's unpaired t test). Each symbol represents an individual mouse.

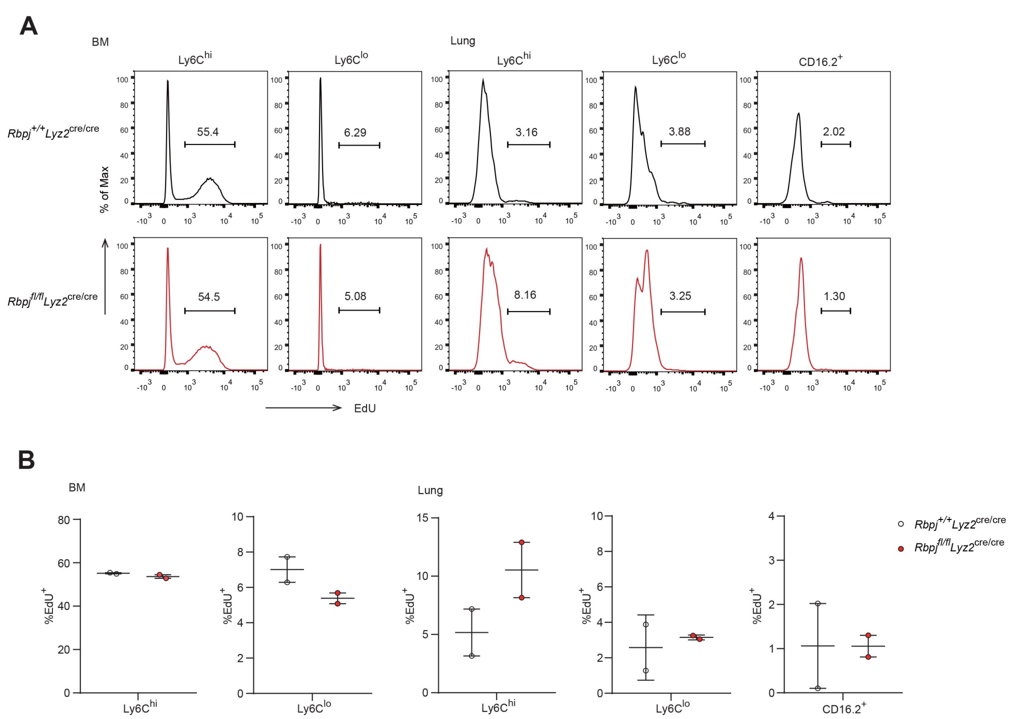
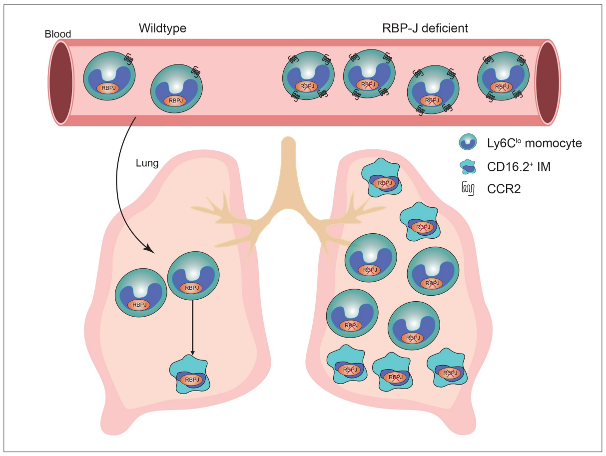


Figure 6 - Figure supplement 1.

RBP-J is not required for turnover of lung Ly6C^{lo} monocytes and CD16.2^{+} IM.

(A and B) Incorporation of EdU was assessed 24 hours after injection. BM monocyte subsets were used as controls. FACS plots (A) and representative data (B) are shown. Each symbol represents an individual mouse.

Supplementary Figure 7



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Reviewer #1 (Public Review):

Kou and Kang et al. investigated the role of Notch-RBP-J signaling in regulating monocyte homeostasis. Specifically, they examined how a conditional knockout of Rbpj expression in monocytes through a Rbpjfl/fl Lyz2cre/cre mouse affects the homeostasis of Ly6Chi versus Ly6Clo monocytes. They discovered that Rbpj deficiency did not affect the percentage of Ly6Chi monocytes but instead, led to an accumulation of Ly6Clo monocytes in the peripheral blood. Using a comprehensive number of in vivo techniques to investigate the origin of this increase, the authors revealed that the accumulation of Rbpj deficient Ly6Clo monocytes was not due to an increase in bone marrow egress and that this defect was cell intrinsic. However, EdU-labelling and apoptosis assays revealed that this defect was not due to an increase in proliferation nor conversion of Ly6Chi to Ly6Clo monocytes. Interestingly, it was revealed that Rbpj deficient Ly6Clo monocytes had increased expression of CCR2 and ablation of CCR2 expression reversed the accumulation of these cells in the periphery. Lastly, they discovered that Rbpj deficiency also led to downstream effects such as an accumulation of Ly6Clo monocytes in the lung tissue and increased CD16.2+ interstitial macrophages, presumably due to dysregulated monocyte differentiation and function.

Their findings are interesting and highlight a previously unexplored mechanistic link between Notch-RBP-J signaling and CCR2 expression in monocyte homeostasis, providing further insight into the distinct pathways that regulate Ly6Chi vs Ly6Clo monocyte subsets individually.

The conclusions of this paper are mostly well substantiated from the experimental data. The strengths of this paper include the use of multiple conditional genetic knock out mouse models to explore their hypothesis and the use of sophisticated in vivo techniques to study the major mechanisms involved in monocyte homeostasis.

Reviewer #2 (Public Review):

The authors provide compelling data to demonstrate that the Notch-related transcription factor RBP-J can influence the number of circulating and recruited monocytes. The authors first delete the Rbpj gene in the myeloid lineage (Lyz2) and show that, as a proportion, only Ly6Clo monocytes are increased in the blood. The authors then attempted to identify why these cells were increased but ruled out proliferation or reduced apoptosis. Next, they investigated the gene signature of Rbpj null monocytes using RNA-sequencing and identified elevated Ccr2 as a defining feature. Crossing the Rbpj null mice to Ccr2 null mice showed reduced numbers of Ly6Clo monocytes compared with Rbpj null alone. Finally, the authors identify that an increased burden of blood Ly6Clo monocytes is correlated with increased lung recruitment and expansion of lung interstitial macrophages.

The main conclusion of the authors, that there is a 'cell intrinsic requirement of RBP-J for controlling blood Ly6CloCCR2hi monocytes' is strongly supported by the data. However, other claims and aspects of the study require clarification and further analysis of the data generated.

Strengths
The paper is well written and structured logically. The major strength of this study is the multiple technically challenging methods used to reinforce the main finding (e.g. parabiosis, adoptive transfer). The finding reinforces the fact that we still know little about how immune cell subsets are maintained in situ, and this study opens the way for novel future work. Importantly, the authors have generated an RNA-sequencing dataset that will prove

invaluable for identifying the mechanism - they have promised public access to this data via GEO.

Weaknesses - The main weakness of the study, is that although the main result is solidly supported, as written it is mostly descriptive in nature. For instance, there is no given mechanism by which RBP-J increases Ly6Clo monocytes. The authors conclude this is dependent on CCR2, however CCR2 deletion has a global effect on monocyte numbers and importantly in this study, it does not remove the Ly6Clo bias of cell proportions, if anything it seems to enhance the difference between the ly6C low and high populations in Rbpj null mice (figure 5C). This oversight in data interpretation likely occurred because this experiment is missing a potentially important control (Lyz2cre/cre Ccr2RFP/RFP or RBP-J variations). In general, there seemed to be a focus on the Ly6C low cells, where the mechanism may be more identifiable in their precursors - likely the Ly6C high monocytes.

Other specific weaknesses were identified:

1. The confirmation of knockout in supplemental figure 1A shows only a two third knockdown when this should be almost totally gone. Perhaps poor primer design, cell sorting error or low Cre penetrance is to blame, but this is below the standard one would expect from a knockout.

2. Many figures (e.g. 1A) only show proportional data (%) when the addition of cell numbers would also be informative

3. Many figures only have an n of 1 or 2 (e.g. 2B, 2C)

4. Sometimes strong statements were based on the lack of statistical significance, when more n number could have changed the interpretation (e.g. 2G, 3E)

5. There is incomplete analysis (e.g. Network analysis) and interpretation of RNA-sequencing results (figure 4), the difference between the genotypes in both monocyte subsets would provide a more complete picture and potentially reveal mechanisms

6. The experiments in Figures 5 and 7 are missing a control (Lyz2cre/cre Ccr2RFP/RFP or the Rbpj+/+ versions) and may have been misinterpreted. For example if the control (RBP-J WT, CCR2 KO) was used then it would almost certainly show falling Ly6C low numbers compared to RBP-J WT CCR2 WT, but RBP-J KO CCR2 KO would still have more Ly6c low monocytes than RBP-J WT, CCR2 KO - meaning that the RBP-J function is independent of CCR2. I.e. Ly6c low numbers are mostly dependent on CCR2 but this is irrespective of RBP-J.

7. Figure 6 was difficult to interpret because of the lack of shown gating strategy. This reviewer assumes that alveolar macrophages were gated out of analysis

8. The statements around Figure 7 are not completely supported by the evidence, i) a significant proportion of CD16.2+ cells were CCR2 independent and therefore potentially not all recently derived from monocytes, and ii) there is nothing to suggest that the source was not Ly6C high monocytes that differentiated - the manuscript in general seems to miss the point that the source of the Ly6C low cells is almost certainly the Ly6C high monocytes - which further emphasises the importance of both cells in the sequencing analysis

9. The authors did not refer to or cite a similar 2020 study that also investigated myeloid deletion of Rbpj (Qin et al. 2020 - 2023 eLife. https://doi.org/10.1096/fj.201903086RR). Qin et al identified that Ly6Clo alveolar macrophages were decreased in this model - it is intriguing to synthesise these two studies and hypothesise that the ly6c low monocytes steal the lung niche, but this was not discussed

Reviewer #3 (Public Review):

In this study, the authors investigate the role of the Notch signalling regulator RBP-J on Ly6Clow monocyte biology starting with the observation that RBP-J-deficient mice have increased circulating Ly6low monocytes. Using myeloid specific conditional mouse models, the authors investigate how RBP-J deficiency effects circulating monocytes and lung interstitial macrophages.

A major strength of this study is that it describes RBP-J as a novel, critical factor regulating Ly6Clow monocyte cell frequency in the blood. The authors demonstrate that RBP-J deficiency leads to increased Ly6Clow monocytes in the blood and lung and CD16.2+ interstitial macrophages in steady state. The authors use a number of different techniques to confirm this finding including bone marrow transplantation experiments and parabiosis.

There are several critical weaknesses that need to be assessed to improve the manuscript, in summary the data presented in the current manuscript are highly descriptive and without mechanistic insight. The inclusion of more mechanistic insight would greatly improve the manuscript.

The authors begin to explore the potential mechanism underlying why Ly6Clow monocytes are increased in the absence of RBP-J - is it through increased survival, increased conversion from Ly6C+ monocytes, increased proliferation or increased egress from the bone marrow. The majority of the data they present here is negative. Whilst I applaud the authors for including negative data, I think that their exploration into how RBP-J leads to increased monocytes does not go far enough and it is critical to understand the mechanism by which RBP-J increases circulating monocytes. Low n-numbers in multiple figures mean that the claims made are not fully supported.

The current title of the paper "RBP-J regulates homeostasis and function of circulating Ly6Clo monocytes" does not fully reflect the manuscript in its current form - there is no exploration of Ly6Clow monocyte functionality in the paper as it stands.
Given that targeting monocytes and macrophages in a range of inflammatory diseases is an attractive yet elusive therapeutic option, understanding the underlying biology that regulates monocyte biology are critically important. This manuscript has the potential to add to our current knowledge of how Ly6Clow monocyte biology is regulated and potentially opens novel avenues for preferentially enhancing Ly6Clow monocytes without influencing Ly6C+ monocytes. This is an attractive proposition for many inflammatory conditions however, considerably more in-depth analysis is required to understand the role of RBP-J in monocyte biology.